

Composition Protocol

Atomic, private, multi-asset settlement for Canton — demonstrated through African commodity trade finance.

HackCanton Season 3 · Track 1 (RWA & Business Workflows) primary, Track 2 secondary · BitSafe Decentralization challenge (secondary entry) · Team: Revotoken Africa

1. Summary

A **private, atomic, multi-asset settlement primitive** on Canton — a reusable Daml building block that bundles several asset transfers into a single deal that either **all** settles or **none** does, with each party seeing only its own leg. CIP-0056 made single assets portable; the Composition Protocol makes them **combinable**. We demonstrate it through African commodity trade finance: an exporter's collateral, a lender's USDCx, and an oracle's price attestation settle as one all-or-nothing transaction, each party seeing only its own leg.

The protocol is the product. The trade-finance flow is the demo that gives it a face.

2. Problem, and why now

Real financial deals are never one transfer. A repo is collateral + cash + a price check. A bond-plus-CDS is a bond leg + a protection leg + a premium leg. On Canton today, every team building one of these **rebuilds from scratch** the logic for making all legs settle together, atomically, while keeping each leg private to the right parties. There is no standard for "all legs or none" with leg-level visibility.

Why now: Canton's institutional volume is real (multi-trillion monthly settlement), CIP-0056 assets are live (USDCx, CBTC, cETH), and the network's economics reward real recurring utility over speculation. The composition layer between tokenization and structured finance is the visible gap — and it is unbuilt.

3. Who it's for (ICP)

Primary customer — builders on Canton

- DEXs, lending protocols and funds (Temple Digital, Tradecraft, Ekiden, Cantex-style venues) that must settle structured, multi-leg deals but don't want to hand-build atomicity and per-leg privacy each time.
- They call the protocol as a shared primitive — the source of recurring, fee-generating on-chain activity that HackCanton's rubric rewards. Specific pain: today they fork-and-rebuild composition logic per product, carrying the privacy and atomicity risk themselves.

Demo persona — African commodity exporters and their financiers

- A commodity exporter with a confirmed purchase order but no working capital; a lender supplying USDCx; a price/grade oracle. They need funding to settle against verified collateral without exposing deal terms to competitors.
- This gives the primitive a vivid, real-world face and a Track 1 end-to-end workflow (create → update → transfer/fulfil → audit) — without collapsing into a single-vertical app.

4. Validation plan (Metrics criterion — do this in-window)

HackCanton scores Metrics/Validation on **evidence from user research, interviews and tests**. This is the current weak point and the plan closes it:

- Interview 3–5 Canton builders (DEX/lending teams in the AppsFactory Discord) on whether they'd call a shared composition primitive — capture verbatim quotes for the pitch.
- Interview 1–2 trade-finance operators (or use documented AfDB/Afreximbank sources) to validate the demo persona's pain — funding delay, deal-term exposure.
- Instrument the MVP: count compositions settled, legs per composition, settlement success/revert rate on the shared DevNet node — these are the on-chain metrics judges want.
- Log all of it in the AI-guided project journal daily (Value → ICP → Metrics → GTM → MVP → Pitch) — the journal is judged and doubles as the traction signal.

5. Why it wins — mapped to the six judging criteria

Criterion	How the Composition Protocol scores
Value / Problem	Names a real, unbuilt gap — the composition layer between CIP-0056 tokenization and structured finance. Why now: institutional volume + live CIP-0056 assets + utility-driven economics.
ICP / Audience	Dual and specific: builders (who call it) + trade-finance operators (the demo). Concrete pain for each.
Metrics / Validation	Builder interviews + operator validation + on-chain metrics (compositions settled, revert rate). See §4.
GTM	Open-source primitive adopted by Canton venues; land with one topology, expand to every structured product. See §9.
MVP	Four Daml contracts, passing tests, Ledger API v2 bridge, live on the shared DevNet node; per-leg privacy proven, not claimed.
Pitch	Tight problem→solution→why-us, with the observer-cannot-see-legs demo as the visible proof.

Differentiator vs. Rocky Exchange, Umbra, Vanton: they build products; we build the primitive they would all call. Composition sits underneath their entire category — we don't compete with Canton's DEXs, we make them composable.

6. Why only Canton

On a transparent chain, private multi-leg composition needs a custom zero-knowledge circuit per asset-type × privacy configuration — complexity and cost scale together. On Canton it takes **two keywords: signatory and observer**. The ledger is the access-control model; privacy is **declared, not bolted on**.

Per-leg privacy is the core claim. Each leg has its own privacy boundary: the lender sees the collateral and cash legs but not the oracle attestation; the regulator observes the settlement receipt but is cryptographically excluded from every leg's payload. Enforced by Canton's stakeholder model — proven by `testAuditorCannotSeeLegs`, not a UI filter. Known counterparties need selective disclosure, not a custom proof system per asset; that is why signatory + observer beats ZK here.

7. Product scope — MVP vs. vision

Ambitious vision, honest MVP. The vision is ecosystem infrastructure every structured-finance app on Canton calls. The MVP is deliberately bounded so it ships, runs, and demos flawlessly — honest scope that works beats broad design that can't be reproduced.

Built for the MVP (in-window, judged)	Vision / cut for now (disclosed, roadmapped)
4 Daml contracts + ComposableAsset interface	Real CIP-0056 registry (mocked at boundary in MVP)
Atomic Settle — all-or-nothing	Heterogeneous leg types via Daml interfaces
Per-leg privacy (observer scoping), tested	Production Allocation-based locking (known upgrade path)
Ledger API v2 bridge; create/exercise/query	Oracle + fiat settlement (mocked in MVP)
3 role views: proposer, counterparty, observer	Governed/threshold composition (BitSafe entry — see §10)
Live on shared HackCanton DevNet node; CBTC/cETH/USDCx legs	Multi-topology library (repo, bond+CDS, fund subscription)

Pre-existing code disclosure: prior Daml contracts and tests predating Sept 18 are disclosed in the README; all delivery-phase work (Sept 18 – Oct 9) is tagged and separately identifiable, per the rules. Judges evaluate only in-window work.

8. The demo that wins

Under five minutes (rule max), one clean topology, privacy made visible — not described:

- Propose a three-leg trade-finance composition (collateral + USDCx + attestation).
- Counterparties accept; the AcceptanceTracker fills.
- Settle — one atomic transaction; all legs commit together. Show the SettlementReceipt.
- MONEY SHOT: observer/regulator view beside a participant view — the observer's screen is provably EMPTY where the legs should be (visibleTokens: []) while the receipt is present. Same ledger, different party, different visibility. Ledger-enforced.
- Atomic-revert case: one leg fails → the whole transaction reverts. No half-state.

9. Go-to-market

- Land: one topology (trade-finance / treasury repo) with one pilot partner on TestNet — 50 compositions, zero settlement failures.
- Expand: second asset class (bond + CDS, heterogeneous legs) — same engine, new leg list.
- Open: publish as open-source infrastructure; distribution is adoption by Canton venues (Temple, Tradecraft, Ekiden, Cantex). The win condition — an outside team builds on the protocol without our help.
- Moat: standards win by adoption, not by license. Being the composition primitive everyone calls is the moat; open source is the point.

10. Prize strategy — track + sponsor challenges

- Primary: Track 1 (RWA & Business Workflows) — the trade-finance workflow is create → update → transfer/fulfil → audit with issuer/holder/observer roles, exactly the track's expected output.
- Secondary: Track 2 (Financial Applications) framing — the protocol drives real economic activity and composability.

- BitSafe Decentralization challenge (50,000 CC, secondary entry): the protocol's Accept step is already multi-party co-signing. Extend it to governed / M-of-N control over settlement (threshold authorization) and enter the contribution pool with a reproducible LocalNet demo — no node required. Natural fit, adjacent work, second prize pool. Only pursue if it doesn't fracture MVP focus.
- Grofty wallet bounty (10,000 CC): SKIP unless the client flow naturally needs a consumer wallet — MainNet-only, real-funds, invitation-gated; not worth contorting an infrastructure primitive.

Draft for team review. Market figures and sponsor claims require source validation. Confirm against final HackCanton S3 rules (submission Oct 9 23:59 UTC).